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Explainable AI E-commerce Recommender System

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Explainable AI for E-Commerce Recommender Systems

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Copyright.....	3
DECLARATION.....	4
Chapter 1: INTRODUCTION.....	6
1.1 Background.....	6
1.2 Problem Statement and Research Questions.....	7
1.3 Project Objectives.....	8
1.4 Scope.....	8
1.5 Organization of Chapter.....	9
Chapter 2: LITERATURE REVIEW.....	10
2.1 Overview of E-Commerce Recommender Systems.....	10
2.2 Overview of Explainable AI (XAI).....	10
2.2.1 Explainable Artificial Intelligence (XAI).....	10
2.3 Core Techniques in Recommender Systems.....	13
2.3.1 Collaborative Filtering (CF).....	15
2.3.2 Content-Based Filtering (CB).....	17
2.4 Related Work.....	20
2.5 Summary of Related Work.....	28
2.6 Summary Of Chapter.....	32
Chapter 3:Theoretical Framework.....	33
3.1.1 Collaborative Filtering Theory.....	34
3.1.2 Theory of Human-AI Collaboration.....	34
3.1.3 Electronic Consumer Decision Process (eCDP).....	35
3.2 Conceptual Framework.....	36
3.3.1 Independent Variable (IV).....	38
3.3.2 Mediating Variables.....	38
3.3.3 Dependent Variable (DV).....	38
3.4 Research Hypotheses.....	38
3.5 Summary of Chapter.....	39
Reference.....	39

Chapter 1 : Introduction

1.1 Background

E-commerce platforms have significantly expanded over the past few years. They bring millions of products to buyers from all categories. The number of options continues to increase. This leaves users to deal with too much information at once. It is difficult for them to spot items that fit their needs. Recommender systems step in to handle that issue. They form a key part of the current online shopping setup. These tools are used to examine how people behave online. They look at things, such as browsing records, buying habits, and interactions with products. From there, they created tailored suggestions for items. This increases user satisfaction. It also drives up sales numbers.

Standard recommender approaches work well in guessing what users might like. Collaborative filtering or content-based models However, many of these setups act as black boxes. They deliver suggestions without explaining why a specific product is shown. However, this opacity creates problems. Users tend to trust the system to a lesser extent. Debugging what goes wrong inside the model is difficult. Meeting rules on how algorithms should be accountable also becomes increasingly difficult.

Explainable artificial intelligence (XAI) is a solid solution to these drawbacks. These methods work to clarify machine learning models. They offer simple reasons for people to grasp each prediction. Regarding recommenders in e-commerce, this clarity shows users the reasoning behind picks. This might stem from the products they checked out before they were alike. or from what others enjoyed similar tastes. Alternatively, even from standout features of the item itself. This openness builds a stronger trust among users. This sharpens their choice. Over time, this leads to greater involvement and actual purchase.

Bringing explainable AI^[P1] into e-commerce recommendations is important. Businesses hold the precision of their personal touch. They also lifted the transparency, fairness, and faith of the customers. This paper describes the application of XAI techniques to these systems. The aim is to deliver explanations that count on suggesting products in online retail.

1.2 Problem Statement and Research Questions

Modern eCommerce recommendation engines like Deep Learning and Collaborative Filtering are highly predictive; however, they are too "black box" systems that provide no reasoning behind their personalized recommendations. Users do not know how the system decides to recommend a particular product to them, which can lead to many problems. First, without any way to verify the reasoning behind the recommendation, users have very low levels of trust in the system and are not inclined to purchase products from it. Second, because the recommendation engine operates as a "black box," developers and stakeholders find it difficult to identify any bias or error within the model, making it impossible to hold anyone accountable for aberrations in the recommendation process. Finally, nearly all successful metrics measure only the accuracy of the recommendation, not the interpretability or level of confidence a user has about their decision-making process, leading to a need to shift away from the use of opaque algorithmic product suggestions and move towards[P2] more transparent and explainable recommendations for users to increase their engagement with these technologies.

To address these problems, this project aims to answer the following research questions:

1. How can Explainable AI (XAI) techniques be effectively integrated into e-commerce recommendation systems to mitigate the "black box" problem?
2. What is the impact of providing visual or textual explanations on user trust and perceived transparency compared with traditional baseline models?
3. How does the trade-off between predictive accuracy and model interpretability affect the overall performance of a recommender system?

1.3 Project Objectives

The main goal of this project was to develop an Explainable AI-driven recommender system that balances high recommendation quality with user-centric transparency. The specific objectives of this study are as follows:

1. **To develop** an e-commerce recommender system engine (using techniques such as Collaborative Filtering or Hybrid approaches) that integrates XAI methods to generate real-time explanations for product suggestions.
2. **To implement** specific explainability techniques (e.g., feature-based explanations or post-hoc methods such as LIME/SHAP) that clarify *why* a product is recommended based on user history or item attributes.
3. **To evaluate** the system's performance using both computational metrics (e.g., Accuracy, Precision) and user-centric metrics (e.g., Trust, Transparency, and Satisfaction) to determine the effectiveness of the explanations.

1.4 Scope

The scope of this project defines the boundaries of the research and development work:

- **Domain:** This project focuses specifically on **E-Commerce**. It utilizes a standard e-commerce dataset (e.g., Amazon Product Data or similar) containing user ratings, reviews, and product metadata.
- **Target Users:** The system is designed for online consumers who face information overload and require assistance in decision making.

- **Techniques:** The project will focus on **Hybrid** or **Collaborative Filtering** models as the baseline recommender, enhanced by **Post-hoc** or **Model-Intrinsic** XAI techniques.
- **System Deliverable:** The primary output will be a functional prototype of a recommendation engine that displays products alongside their explanations (e.g., "Recommended because you bought X" or "Similar to items you viewed").
- **Limitations:** This project does not involve live commercial deployment or real-time payment processing. The evaluation will be conducted in a simulated environment using historical data, and potentially a user study or offline evaluation metrics.

1.5 Organization of Chapter

The structure of this report is organized as follows:

- **Chapter 1: Introduction** provides the background of the study, outlines the problem of "black box" recommenders, and defines the research objectives and scope.
- **Chapter 2: Literature Review** offers a comprehensive analysis of existing E-Commerce Recommender Systems and Explainable AI (XAI). It reviews core techniques, such as Collaborative and Content-Based Filtering, discusses XAI categories, and summarizes related works regarding user trust and evaluation metrics.
- **Chapter 3: Methodology** (Future work) will detail the system architecture, selection of algorithms, data preprocessing steps, and experimental design proposed to achieve the project objectives.
- **Chapter 4: System Design / Preliminary Results** (future work) will present the initial framework, interface design, and any pilot data analysis.

- **Chapter 5: Conclusion** (future work) summarizes the project's contributions, limitations, and potential for future enhancements.

Chapter 2: LITERATURE REVIEW

2.1 Overview of E-Commerce Recommender Systems

Since the mid-1990s, the explosion of the Internet has exacerbated the problem of information overload among both consumers and producers. Consumers struggle to find products they may want among countless alternatives that may make the task less efficient or time-consuming, whereas producers struggle to give their products visibility in an increasingly crowded online marketplace. Recommender systems (RS) emerged as an effective technological remedy to these problems of imbalanced exposure. These systems connect users with information related to their interests, thereby helping consumers identify items of interest and producers expose their goods to potential buyers in the marketplace to efficiently build a market environment. By the late 1990s, Amazon, eBay, and Alibaba developed personalized recommender systems based on user profiles, purchasing trends, and item popularity-in-the-last-week to create personalized product recommendations.

As global e-commerce continues to expand across various industries, including digital services, retail, and online marketplaces, the limitations of traditional recommendation systems have become increasingly evident. Historically, most RS emphasized predictive accuracy, often overlooking the need for transparency and interpretability. However, many users navigate online platforms without a clear purchasing intention, and some may lack domain knowledge of certain product categories, such as financial instruments or healthcare-related items.

2.2 Overview of Explainable AI (XAI)

2.2.1 Explainable Artificial Intelligence (XAI)

Recently, the use of artificial intelligence (AI) to assist in decision-making has expanded into high-stakes areas, such as the employment screening process, insurance pricing, loan approvals, and even within the criminal justice system. While these algorithms offer an increase in efficiency and predictive abilities, their black-box nature raises reasonable concerns about bias, discrimination, and adverse hostile outcomes. Many articles document how black-box models can inadvertently perpetuate inequality or the dolls/guidelines trained on previous processes, creating bias.[H3]

One such example is personalized news feeds and targeted advertising on many social media platforms. This type of advertising has been scrutinized as a reason why people experience filter bubbles, which decrease the diversity of information being presented to them. Unlike recommender systems- in which a user can at least choose if they want to select one of the suggested options, many algorithmic decision systems provide just one automated decision with no chance for user intervention - exacerbating the negative consequences when the user doesn't understand how the system came to their decision.

Consequently, transparency, user control, and accountability are included when researchers recommend the design of artificial intelligence (AI) systems. The potential for users of an AI model to have transparency, in effect, allows them to discern bias, understand how the AI model made certain predictions, and avoid over-reliance on normative algorithms. Prior research has suggested that intelligent systems should not act autonomously on behalf of the user; rather, the user should remain looped in the decision-making process to preserve accountability during the process. With the increase in legal and ethical matters surrounding algorithmic decision-making, an explainable AI framework has proven to be an applicable foundation towards the fairness, trust, and accountable behavior of machine prediction.

2.2.2 Categories and Techniques of Explainable AI

Techniques for explainability in Artificial Intelligence are used to help users understand how and why a model generates predictions. Explainability techniques are crucial in recommender systems because the degree of transparency can greatly impact trust in the model and the acceptance of decisions. AI can be broadly classified into four categories: model-agnostic, model-specific, intrinsic, and post hoc.

Model-agnostic techniques provide an interpretable framework that explains the predictions of any machine-learning model. **Prominent examples include LIME and SHAP**, which enable explanations regardless of the underlying algorithm (Ribeiro et al., 2016; Lundberg & Lee, 2017). These frameworks are useful in recommender systems because they use local or instance-level explanations and do not affect what the recommender system does algorithmically. For example, in e-commerce systems, these tools serve the user by explaining why products are recommended, which increases transparency, understanding, and trust.

On the other hand, model-specific techniques are organized to handle specific model families. For example, attention (or attention-based) mechanisms in neural-based networks, particularly in sequence-based or session-based recommender systems, identify important user-item interactions in the model (Chen et al., 2019). Model-specific techniques also provide a more accurate explanation because they utilize internal model components such as weights, embeddings, or attention distributions.

Explainability methods are often divided into intrinsic and post hoc methods. Intrinsic explainability is related to models that are interpretable by nature, such as decision trees, rule-based systems, or matrix factorization models with explicit latent features (Melis & Jaakkola, 2018). Intrinsic explainability provides transparency but often results in a reduction in predictive accuracy **compared to complex "black-box" models**. Post-hoc explanations are intended to create explanations after the model has been predicted. Post hoc explanations include feature importance visualizations,

counterfactual explanations, and natural language explanation activities (Guidotti et al., 2018).

In e-commerce recommender systems, post hoc techniques have become popular because they can provide an additional method of interpretability in addition to state-of-the-art **black-box** models such as deep learning recommenders. Studies have shown that explaining recommendations not only builds user trust, but also increases user engagement and conversion by helping users gain verifiability of recommendations (Zhang & Chen, 2020). Therefore, integrating explainability methods into AI-based recommendation engines is becoming a baseline requirement for e-Commerce platforms.

2.3 Core Techniques in Recommender Systems

Modern Recommender Systems (RS) are based on three main types of technology. All of them began as solutions for dealing with the increasing amount of information available to consumers in e-commerce. First, most businesses use Collaborative Filtering (CF), which states that if two people agree about something in the past, they will also agree about it in the future (Goldberg et al., 1992). CF can be broken down into three basic types of classifications: First, User-Based (UB) CF, where we recommend items based on what nearby users with similar tastes like; Second, Item-Based (IB) CF, where we recommend items that are similar to what our users have previously liked (Sarwar et al., 2001); and third model-based (MB) CF, where we use machine learning methods, such as Matrix Factorization and Deep Learning, to learn embeddings (Hidden Features) for better and more complex predictions about what a user will want next. Of these three methods, Item-based CF is the most scalable for large-scale platforms, making it very popular among retailers that provide many products (Linden et al., 2003).

Another big method of recommending all kinds of products and services to users is through a method called Content-Based Filtering. This technique uses the attributes of both items and users to profile the types of items a user would like, based on the attributes (such as categories or descriptions) of items that have already been liked (Pazzani & Billsus, 2007). For example, if a user has purchased books relating to

"Machine Learning," this will allow the algorithm to recommend other books with related tags or topics. Hybrid Recommender Systems combine the advantages of CBF and Collaborative Filtering to improve the results by considering both methods. There are several examples of hybrid recommendation systems, including weighted hybrids, feature combinations, and cascading hybrids (Burke, 2002).

Although the three main technologies, Collaborative Filtering, Content-Based Filtering, and Hybrid Systems, have improved the accuracy of predictions and consequently increased the sales of products through e-commerce sites as shown in Table 1.1. They all have one common flaw. Many of these are black box models, particularly those based on deep learning, which means they can generate very accurate recommendations; however, they do not explain how they arrived at their conclusions in a way that humans can understand (Guidotti et al., 2018). This lack of transparency makes it difficult for users to trust the model, find errors, and comply with accountability for algorithms. Explainable Artificial Intelligence (XAI) was developed to address this significant lack of transparency (Zhang & Chen, 2020).^[H4]

Table 1.1 Key features, advantages and disadvantages of recommender systems^[H5]

Technique	How it Works	Advantages	Disadvantages
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Collaborative Filtering	Recommends items based on the history of similar users or similar items.	1. Serendipity: Can discover the new interests for users 2. Domain Free: No need to understand what the item is. 3. High Accuracy: Provides a very accurate predictions when data is abundant.	1. Cold Start Problem: Cannot recommend to new users or new items. 2.Data Sparsity: Hard to find similarities if users have rated very few items. 3.Black Box: Hard to explain why a similarity exists.
Content based filtering	Recommends items by matching item features with the user's profile	1. User Independence: No need other users data to work 2. No Item Cold Start: Can recommend brand new items as soon as they are uploaded . 3. Transparent: Easy to explain	1.Over-Specialization: Users get trapped in a "Filter Bubble". 2.Limited Novelty: Rarely recommends something surprising or new. 3. Tag Dependency: Requires very accurate and detailed descriptions of items.

Hybrid Recommend System	Combines CF and CB methods to leverage the strengths of both and minimize their weaknesses.	1. Best Performance: Easy to achieve higher accuracy than CF or CB. 2. Solves Cold Start: Can use CB for new items and CF for established ones. 3. Robustness: More stable and reliable across different scenarios.	1. Complexity: Much harder to build, tune, and maintain. 2. Computational Cost: Requires more processing power and time. 3. Hard to Explain: Mixing multiple models makes the logic very complex, leading to the "Black Box" issue.
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2.3.1 Collaborative Filtering (CF)

Collaborative Filtering (CF) is the most basic and widely used method for building Recommender Systems. The CF method is based on collective intelligence, and the concept that one can infer a person's preference for an item by examining the history of other people who have interacted with the same or similar item (Goldberg et al., 1992). The CF algorithm assumes that people who exhibit similar consumption patterns tend to prefer the same items in the future. The CF system is an excellent solution to the problem of information overload in the e-commerce world, helping provide unexpected (but welcomed) product recommendations (Adomavicius & Tuzhilin, 2005).

Collaborative Filtering (CF) exists in two main classifications, one of which is memory-based CF. This type of CF looks at the complete history of user-item interactions to determine how closely related users (User-Based CF) or two items

(Item-Based CF) are by calculating their relationships based on their interaction with many other users/items (Sarwar et al., 2001). In User-Based CF, users who have similar interests as the user being recommended have the highest likelihood of liking the same items, thus providing a recommendation. This method is costly to compute on a large scale because of the sheer volume of data; therefore, it is not as scalable as Item-Based CF in large-scale e-commerce settings. However, Item-Based CF is much more organized and reliable (because it is based on item pairs) than User-based CF and is commonly used in major e-commerce platforms such as Amazon because it provides a better overall user experience and scalability across highly populated systems (Linden et al., 2003).

Model-Based Collaborative Filtering is a second type of collaborative filtering. Using machine learning and data science approaches, a user preference model was created by identifying latent features and hidden structures within the data. A well-established methodology used in this area is Matrix Factorization, which creates a low-dimensional representation of users and produces item representation (Koren et al., 2009). The resulting representations for both the user and item are used to calculate the rating for the user/item pair in this low-dimensional space. The development of modern Model-Based Collaborative Filtering also includes deep learning techniques, such as Neural Collaborative Filtering, which utilizes deep neural networks to model complex nonlinear relationships between user and item embeddings. The result is typically better predictive accuracy compared to classical techniques (Zhang & Chen, 2020).

CF, while effective, has several limitations (many of which also exist with other forms of deep learning) that prompt the need for Explainable AI (XAI). E-commerce sites often experience problems owing to both sparse datasets and the Cold Start Problem, which creates challenges in predicting user behavior on the basis of limited pre-existing interactions (Adomavicius & Tuzhilin, 2005). The primary drawback of Model-Based CF approaches is that they produce high-quality recommendations through a black box model that lacks the transparent logic behind the reasoning used to generate their recommendations (Guidotti et al., 2018). Because of this lack of explanation, the users of CF systems can lose faith in the accuracy of the CF service and thus do not want to make purchases or keep accounts from these

vendors (Tintarev & Masthoff, 2007). This gap between high-quality recommendation services and user trust is what XAI techniques have been created to close (Zhang & Chen, 2020).

2.3.2 Content-Based Filtering (CB)

Content-Based Filtering (CB) is part of a Recommender System approach that suggests a new item based on how similar the items were rated by the user in the past. This is different from CF, which decides what to suggest based on the overall behavior of other users of the system (Pazzani & Billsus, 2007). The basis for CB is merely the history of transactions and likes by a particular user, along with their attributes. CB operate on two fundamental concepts. The first is to build an Item Representation that essentially represents every item in an e-commerce catalog[P6] through a set of specified attributes, such as color, brand, category, and technical details.

The next step is to create a User Profile. What we do here is to add together a User's "explicit" interactions with items (like Ratings) and their "implicit" interactions (such as Clicks or Viewing Time) to create a numerical representation (a Mathematical Vector or Profile) of what the user likes. In essence, the profile is a condensed version of the different types of features that make up a user's favorite items (Pazzani & Billsus, 2007). When the profile is completed, the Recommendation Engine can then calculate the similarity between the user's profile and all other un-interacted items in the catalog (separate from everything else). The items with the highest degree of similarity (meaning they have many features in common with what the user has historically chosen) are then presented to the user as recommendations. For example, if a user consistently clicks on or buys accessories for 'Smart Home' technology, the algorithm will recommend new products from the 'Smart Home' category and may even include products from the 'IoT' category.

With Content-Based filtering, it is easier to understand how the system generates recommendations based on the features of the user profile and those of the items being recommended. Recommendations are created by matching the features of the item with those of the user's profile. In other words, it allows for straightforward

explanations such as, “We recommend this product because it has the same features found in the products you previously purchased” (Tintarev & Masthoff, 2007). Because of how clear and straightforward the reason for the recommendation is, this makes the system much more trustworthy than non-transparent CF systems. In addition, because the recommendation is based solely on a complete description of the features of an item, CB filtering systems can easily recommend new items that have yet to receive any user input, which solves the cold-start problem with respect to new products.

Content-Based Filtering (CB) certainly has limitations, which include Over-Specialization and Feature Dependence (Zhang & Chen, 2020). Users can easily become trapped in a filter bubble because of the CB’s tendency to recommend only products that are almost identical to what they have liked before; thus, there is no chance that they will stumble onto something new, unique, or surprising. The performance of CB is completely determined by how good, complete, and uniform features are recommended for each product. If a product does not lend itself to easy labelling with explicit descriptors, the CB will not perform well. While these are obviously two shortcomings of CB, the way that CB is structured allows it to be incorporated easily into Hybrid Recommender System designs (Burke, 2002), and as a firm foundation for developing advanced explainable AI applications.

2.3.3 Hybrid Recommender System

Hybrid recommender systems represent the third main type of technology for creating recommendations designed to minimize the limitations of both Collaborative Filtering (CF) and Content-Based Filtering (CB), while leveraging the strengths of both techniques (Burke, 2002). CF has traditionally struggled with the cold-start problem for new users and offers little understanding or interpretability. However, content-based filtering often suffers from a lack of serendipity and the ability to help users discover new items outside their existing profile. Hybrid Recommender Systems aim to combine the advantages of both approaches to create an optimal blend of predictive accuracy, coverage, novelty, and ability to provide users with robust explanations for recommendations. The current trend in e-commerce is to

utilize hybrid systems to provide high-quality recommendations that adapt to constant changes in product catalog and customer bases.

There are various architectural strategies for creating hybrid systems, each of which provides a unique mechanism for combining source systems. The most prominent strategy was the Weighted Hybrid method. In this structure, the scores or predictions generated by different CF and CB systems are linearly combined using predetermined weights (Burke, 2002). Another hybrid structure is Feature Combination, where the output or model components from one system are used as input features into another system. Finally, the Cascading Hybrid allows sequential building. One system acts as a coarse filter that limits the number of items to choose from, whereas the other, usually more sophisticated, takes the reduced candidate field to perform the final ranking and selection (Burke, 2002).

The adoption of hybrid architectures in e-commerce has shifted how businesses approach product personalization. These architectures enable companies to integrate multiple models to generate superior predictions for user actions. By considering diverse data types and user-item interactions, Hybrid Systems can deliver accurate recommendations based on unique characteristics for established users, while simultaneously handling new users (cold start) through content features. Furthermore, by utilizing multiple algorithms, users are less likely to fall into the "filter bubble" often associated with pure CB systems, gaining access to a greater variety of products and increased novelty (Zhang & Chen, 2020).

The integration of Explainable AI (XAI) is particularly critical in Hybrid Recommender Systems (HRS). The HRS helps to find a balance between predictive accuracy and explainability. A common example involves utilizing a CF component to generate accurate (but opaque) predictions, while simultaneously using a CB component to generate the rationale behind the recommendation (Tintarev & Masthoff, 2007). Although the problem of providing accurate explanations becomes more difficult as recommendations are derived from complex, multimodel interactions, a well-designed Hybrid Recommender System can provide a high-performance solution that meets the demands of user trust and algorithmic accountability (Zhang & Chen, 2020).^[H7]

2.4 Related Work

1.Stability of Explainable Recommendation

Vijayaraghavan and Mohapatra (2023) stated that one of the most significant areas of research regarding explainable recommender systems is the reliability of the explanation given by the recommender and how it impacts the user's understanding of how to interpret their recommendations. It is known that many users rely on their understanding of why a recommendation has been made when determining whether to accept it and, consequently, how to act in relation to it. However, there has not yet been any rigorous investigation of the impact of different scenarios on the reliability of the explanation associated with a recommender and, therefore, its ability to provide accurate information to the user. This study seeks to address the issue of the risk of unreliable explanations resulting in a potential threat to the user by providing empirical evidence demonstrating the susceptibility of current feature-based explainable recommendation systems to external noise and malicious attacks.

To determine the stability of the explanation for perturbations of the model's parameters, we used a white-box attack methodology as the basis for our experimental design. Our experimental design relied on two different types of noise injection: Random Noise and Adversarial Noise. Random Noise was created by adding Gaussian noise to the model parameters. Adversarial Noise was created by optimizing the perturbation directly against the model's objective function using the Fast Gradient Sign Method (FGSM). This empirical experiment was intended to assess how adding different amounts of external noise degrades the performance and explanation quality of three cutting-edge feature-based explainable recommender systems.

Two popular ecommerce datasets, Amazon Electronic and Yelp, were used by the investigators to evaluate the models. The investigators provided an empirical confirmation of their research hypothesis regarding how much the ability of a

recommender system to give explanations fades away as noise levels being introduced to a listing increase tremendously. This investigation has shown that all tested models for giving explanations are susceptible to dissecting wrong detentions created by some kind of "adversarial" or "misleading" inputs rather than "crazy" or randomly introduced variations of data on listing items. This empirical validation adds to the pressure to create effective methods for enhancing the reliability of e-commerce explainable recommender systems (ERS), with the ultimate goal of increasing the trust consumers have in these types of systems.

2. Exploring Customer Behavior and Revenue Generation via Explainable Artificial Intelligence (XAI) in E-Commerce

According to Azad (2023), the main issue with machine learning algorithms in e-commerce is that while many of them are quite accurate, their low level of user trust leads to problems with usage. Without an explanation as to why a model predicted an output, for instance, managers and businesses lack insight into their customers' buying behaviours. In his study, Azad found that machine learning with explainability (XAI) provide the ability to see into the specific factors influencing a model's recommendation to help build trust and accountability. Lastly, he discussed how regulations, such as the General Data Protection Regulation (GDPR), do create additional requirements for transparency when classifying data for a digital business.

The CRISP-DM process was followed to implement a working XAI model for eCommerce use cases. To create the primary predictive engine, the investigator chose to use Random Forest and ANN algorithms, and trained on Kaggle's dataset of 13,000 records. Three separate methods were used to achieve interpretability by integrating LIME, SHAP, and Eli5 frameworks. The methodology used for creating an XAI was made up of five stages: preparing, exploring, building, evaluating, and employing XAI Frameworks to obtain local and global explanations.

The integration of XAI enabled e-commerce retailers to accurately identify which product features were most responsible for increasing customers' likelihood of making purchases . One of the key benefits of XAI is the ability to maintain the level of performance (accuracy) of a complex ANN while providing the retailer with a

human interpretable rationale for how and why consumers are making decisions. The authors found that SHAP provided an equitable distribution of feature contributions across the entire model, based on concepts from game theory, thus increasing the trustworthiness of the SHAP explanation. Additionally, XAI was demonstrated to support e-commerce business strategy, for example in pricing strategies and marketing strategies/sales.

Although the author describes several benefits of the framework described in his paper, he also notes many of the limitations, specifically the inherent trade-off between accuracy and interpretability and the assumption that the training data was of good enough quality; the depth of noise and quality was not explored in detail. Additionally, the author noted that the exclusion of discrimination (race, age, gender, etc.) would promote fairness but may have limited the overall predictive power of the models. Lastly, some frameworks (specifically, ELI5) performed poorly in explaining more complex black boxes in comparison to LIME and SHAP.

By employing the confusion matrix to evaluate the performance of the models, the researcher also looked at three separate measures; Precision, Recall, and AUC scores. The researcher also performed manual inspections of the SHAP Summary Plots and Waterfall Charts in order to ascertain that the explanations of the models were in accordance with real-life business reasoning. It was found that LIME was better at providing accurate local explanations for single instances, however, SHAP provides an assurance that interpretability is met through the axioms that govern the method. In conclusion, this study found that ANN combined with SHAP produced the most accurate predictions as well as the highest quality of explanation.

3.Explainable AI-Driven Recommender Systems in E-Commerce

In the paper written by the author, Rahman and (2024), they discussed [H8] the long-standing issued with "black boxes" in e-commerce recommendation systems. The black-box effect of AI means that users cannot see or understand how

recommendations are created. This can be an obstacle when users make trusting decisions. To reduce user distrust, this research looks at how to increase users' trust and perception of transparency and its impact on actual behavior by implementing Explainable AI (XAI) features such as concise and contextual explanations.

The study's methodology integrated quantitative and qualitative mixed-methods approaches. Quantitative data accounting for causal relationships were obtained from findings based on an online between-subject experiment involving 450 individuals, whereas qualitative perceptions of the user regarding the recommender system were obtained through semi-structured interviews (30).

Random assignment assigned subjects to view either the regular version of the algorithm-based recommender system or an experimental version of the system enhanced with XAI. Participants in the XAI condition were given dynamic and succinct explanations of the underlying logic used to generate recommendations.

The effects of the experimental system on user confidence were examined through a series of statistical analyses. The analysis showed that users who interacted with the XAI-enhanced system had greater levels of trust ($M=4.1$ vs. 3.2) and greater levels of perceived transparency ($M=4.3$ vs. 2.9) compared to users who interacted with the traditional control group. Most importantly, the mediation analysis indicated that trust and perceived transparency were fully mediating variables in the link between XAI exposure and decision to purchase. This suggestion has proved that explanation recommendations through XAI have created a high reliability for users.

4. Machine Learning Based Potential Customer Recommendation System for E-Commerce Using Feature Selection and XAI

The authors, Peya et al. (2025), pointed out that with digital commerce growing increasingly complicated due to the huge number of consumers, merchants face an increased level of decision fatigue as well. The study emphasized that the main component of improving a user's experience and improving sales was through the

use of personalized recommendation systems. They further added that while machine learning was an effective tool to analyze large amounts of historical data, merchants must also understand why certain recommendations were made to create trust among customers. Therefore, the authors wanted to create a "smart" customer recommendation system that utilized Explainable Artificial Intelligence (XAI) in order to support data-driven decision-making.

A customer behavior dataset from Kaggle was utilized for the investigation, which includes 3,900 transaction records. The researchers developed a new methodology for feature selection by applying the Select K-best with Mutual Information (MI) algorithm to identify the impact of the variables selected as important. Classification was performed using and comparing three algorithms: Random Forest (RF), Logistic Regression (LR), and K-Nearest Neighbor (KNN). To add further interpretability, a Local Interpretable Model-agnostic Explanations (LIME) framework was used to explain responsible behaviors and transactions.

According to the findings of this study, the methodology for selecting the best characteristics had a significantly significant effect on increasing the accuracy of the recommendation results and that the Random Forest Algorithm provided a perfect score in accuracy at 100%. A significant benefit of the method was the availability of the LIME visuals, which provide the merchant with the actual weights of variables such as "Review Rating" and "Purchase Amount". The results showed that the visibility of the variables through LIME increased the merchant's confidence in the automated recommendations; therefore, they were more likely to adopt the automated recommendations rather than dismiss them as "black-box" outputs. The modular design of the system made it flexible and able to fit various types of data sets for different e-commerce businesses.

Despite its strong accuracy levels, the researchers' study concluded that various other algorithms, including Logistic Regression, produced results with far fewer performance metrics when compared to ensemble algorithms. The authors believe that the systems' abilities were primarily dependent on establishing logical thresholds, which may include a minimum of \$70 spent or four-star ratings, though these thresholds may change for different types of industries. Additionally, KNN's

accuracy fluctuates more through 5-fold cross-validation when compared to using a 'hold-out' approach. The study focused on discovering potential customers rather than evaluating the quality of the data itself.

The evaluation of this system using a comprehensive set of measures, including precision, recall, F1 score, accuracy and ROC-AUC, demonstrated that the Random Forest model achieved a perfect ROC-AUC score of 100% on both hold-out and 5-fold cross-validation datasets. The researchers performed a comparative analysis against previously published studies, confirming that the Select K-best with MI method outperformed the conventional PCA and FP-Growth approaches. They also assessed the effectiveness of the LIME module to generate stable explanations for individual testing instances, determining that the "Purchase Amount" variable had the greatest impact on predicting the classification of potential customers.

5. Personalized Fashion Recommendation with Visual Explanations

Fashion recommendation systems typically rely on ratings text to generate recommendations. However, they often overlook the visual elements consumers consider when making purchases. Chen et al. (2020) created a multimodal^[P9] attention neural architecture to generate personal fashion recommendations with visual explanations. These visual explanations identify the specific parts of a product's image that correspond to the user's preferences.

This method uses a deep learning model that incorporates both text and imagery into the analysis. This method takes advantage of multiple forms of attention through a multimodal attention neural network that processes regional-level features extracted with convolutional neural networks (CNNs) along with user reviews in order to correlate which regions of an image are most relevant to the training users' historical preferences, and determines the Region of Interest (ROI) that will contribute most substantially to the recommendation score based on the image content alone.

Large-scale fashion data were used by the authors to evaluate the system, and they probably evaluated the quality of the visual explanation using quantitative metrics as well as a qualitative user study to determine if the regions highlighted were relevant to the user, using crowdsourced resources such as Amazon Mechanical Turk. The results indicated that visual attention is useful in improving recommendations, and it allows for a better understanding of the relevant features of a product owing to the natural, interpretable effect of visual attention on users.

6.Explainable AI for Seller Recommendations (XAI-SR)

Akhtar (2024) proposed a mixed-model recommender system to make the seller's recommendation more transparent in the e-commerce marketplace. Traditional recommender systems such as Collaborative Filtering recommend products based on relationships; however, because they cannot explain what highlights a certain seller versus another, they tend to be "black boxes." The potential for eroding trust exists in this scenario as well, especially when seller reputation is a significant factor and seller social connections are also included. Thus, the study's development of a model using Explainable AI (XAI) is intended to provide more personalized recommendations for sellers that produce accurate recommendations, as well as to interpret seller reasons.

Using a mixture of Matrix Factorization and Graph Neural Networks, the system is trained using MF to discover users' latent preferences and understand the complex interactions between users and sellers. The explainable nature of the model is enhanced through the use of two cutting-edge Explainable AI (XAI) models, Integrated Gradients and DeepLIFT, with both having slightly different means of achieving the goal of explainable feature importance and visual representation of recommendations. Integrated Gradients track all the gradients in a model along a path from the baseline and provide an aggregation of all the gradients as the answer, whereas DeepLIFT calculates the importance of individual features to output using their relationship to their baseline input. This allows for an explanation of the specific recommendations and the reasoning behind them.

The model's performance and explainability were evaluated based on e-commerce datasets used in the real world. The hybrid approach exhibited greater accuracy than either MF or GNN alone. Additionally, through a user feedback survey, we discovered that 82% of the respondents said they trusted the explanations given along with the results and 76% felt confident when making a choice about what to buy after receiving an explanation.

This strategy has two levels of explainability: a high level and an instance-by-instance level. This allows users to view the workings and reasons for recommendations of the system at two distinct levels, thereby earning their confidence and engagement, two critical factors for ensuring loyalty to the platform.

However, the model is expected to face challenges with the computational complexity of running more sophisticated XAI solutions such as Integrated Gradients, which require significant hardware resources to be used in real time for every recommendation made, especially in high-volume environments. Moreover, while the current architecture has a stronger orientation for using a large amount of structured data for generating explanations than it does with unstructured multimodal data (e.g., seller images, detailed written reviews), this may limit its applicability to these types of data.

7.An E-Commerce Recommendation System Based on Consumer Behavior Models

The Pleskach et al. (2023) research provided the initial framework for a new type of recommendation engine called the "Multivariate Approach to Recommendations." The Multivariate Approach integrates the most recent findings of consumer behavior theory and a micro-service architecture to provide contextualized recommendations.

Numerous traditional recommendations do not use a multitude of variables such as geographic location, environmental conditions (such as outside temperature), and timing that affect a consumer's purchasing decision. The research team proposed merging consumer behavioral analysis with modern technical implementations to fill

a void in how e-commerce recommendations are currently implemented and create a more contextualized and personalized shopping experience for consumers.

In their methodology, the authors adopt a dual approach that consists of a theoretical model and technical architecture for their proposed electronic store. The theoretical aspect of this study uses the eCDP model as the foundation. The authors adapt the "problem recognition" stage of the eCDP by incorporating an anonymous user pseudo-identification mechanism that allows for dynamic adaptation to the interface. The technical aspect of the methodology employs a microservice architecture for an electronic store, which includes separate application programming interfaces (APIs) for order management, cart services, and a specific recommender system package. The recommender system package comprises a chatbot, knowledge base, and statistical records, which provide an interface to process customer requests and generate real-time recommendations based on enriched customer data such as location and weather conditions.

This research assessed the usefulness of the suggested behavioral elements by conducting an extensive examination of e-commerce purchases throughout Ukraine based on traditional geographic boundaries (regions). Average purchase value (APVs), purchase frequency rate (PFRs), and customer value (CVs) are considered important measurements of e-commerce purchases in this region. Regional differences were evident in the results obtained from the analyses (i.e., Odesa had the highest Customer Lifetime Value). Additionally, outside influences had a measurable positive impact on sales. For example, engaging with customers on rainy days increased children's participation by 14.2%. These quantitative results confirmed that it is beneficial to consider external factors when determining the method used to suggest products.

One of the main advantages of this system is that it offers context-aware and context-responsive functionality. This feature allows for the provision of highly applicable recommendations by considering not only past purchases but also other external triggers such as weather patterns and regional economic indicators. The microservice-built architecture also allows for added scalability and modularity by allowing for independent scaling or updating of parts of the system without disrupting

the entire solution. Furthermore, using a chatbot for real-time interaction with users reinforces the notion of "explainability" and "engagement" with the recommendations.

Nevertheless, this research also identifies limitations in terms of data complexity and the cost associated with system implementation. The creation of a fine-grained, context-sensitive system necessitates the gathering and analysis of large quantities and types of weather data, location data, and end-user actions, which create significant expenses through computation as well as associated privacy issues. Furthermore, although the results from the regionalized study yielded some useful insights for Ukraine, the specific behavior results may not be fully transferrable to other nations with distinct cultural/economic conditions until such behaviors are localized and tested against multiple locations.

8.Human-AI Collaboration in Explainable Recommender Systems

Waykar (2024) insists that the worth of working with Humans and Artificial Intelligence to develop Explainable Recommender Systems (XRS) is the most important one. Traditional AI-based recommenders can be successful at recommending content, but the majority of the time, they are "black box" solutions. Consequently, users have no insight into the rationale behind specific recommendations. Because of this lack of knowledge, users are often distrustful or skeptical of such systems, and the views provided by the algorithm are generally viewed as less accurate than they might otherwise be. The goal of this study was to create a better understanding of the relationship between algorithmic accuracy and user comprehension by evaluating how users utilize AI-generated explanations. In addition to looking at how users utilize AI-generated justifications, this study also focuses on ways to increase the transparency of these explanations, making them more understandable to users. This paper seeks to find a way to create a system

that delivers both accuracy and transparency through effective collaboration between user cognitive processes and AI-generated justifications.

The research methodology includes an extensive review of different techniques intended to support collaborative efforts between humans and AI. This research focuses on three major areas of explanation methods: model-based, post-hoc, and interactive interfaces. Additionally, this study provides methods for users to create explanations that reflect their individual preferences, such as choosing between different types of explanation formats and obtaining contextual feedback to refine how AI explains its recommendations using fine-tuning techniques. This is a much more comprehensive way of empowering users by allowing them to participate in the recommendation process instead of simply receiving a predetermined recommendation from an artificial intelligence system.

An Evaluation Framework has been established to evaluate the human–AI collaboration system, which can effectively determine the effectiveness of a human–AI collaboration system. The Evaluation Framework goes beyond traditional accuracy measurement and includes a number of additional user-focused metrics, such as Explanation Quality; User Understanding; Trust & Satisfaction; User Engagement.

The Evaluation Framework not only compares the collaborative XRS against traditional black box systems with the aim of determining the difference in user confidence in the decision-making process over time but also helps to identify the strengths and weaknesses of current collaborative XRS systems and provide a roadmap for future improvements.

One key benefit of this collaborative model is that users gain greater confidence in AI and feel empowered by their ability to support decision-making processes. Users will have direct access to and the opportunity to manipulate AI's language and explanation style, thus respecting individual cognitive variances. Furthermore, since both new and experienced users will be able to utilize various types of explanations, AI will accommodate users with different backgrounds, as well as continue learning

from user feedback. Therefore, as they interact with AI, users will receive increasingly tailor-made and pertinent recommendations based on their experience.

However, this study has important limitations and challenges. Collaborations require balancing the trade-off between providing simple and thorough explanations. Providing overly extensive or detailed explanations can create confusion for users, whereas providing too many simplistic explanations can create misunderstandings. There is also a technical concern regarding whether these types of explanations affect the predictive quality of the system or expose sensitive information about users. Another issue cited in this paper is that there are no standardized metrics for evaluating the quality of the explanation; therefore, it is difficult to accurately evaluate the quality of the explanation from a user perspective. In addition, because of cognitive biases, users continue to misinterpret explanations, even when they have objective evidence (and a detailed explanation) about the output obtained.

2.5 Summary of Related Work

This section shows the overviews of the previous related work. Table 2.1 provides a summary of the key aspects and findings of the field's research.

Table 1.2 Summary of Key Aspects and Findings

References	Paper Title	Key Findings	Evaluation Metrics	Datasets
(Chen et al.,2020)	Personalized Fashion Recommendation with Visual Explanations	A multimodal neural network was created by the Author which takes into account the user's review text combined with visual features to create personalized fashion recommendations for the user. The results showed that integrating	Recommendation Performance: Although specific names (like NDCG or AUC) were not listed in the text, the authors used quantitative metrics to measure if the system successfully	Large scale Fashion Data.

		visual representation into the recommendation system allows for greater accuracy in recommendation generation while enhancing the interpretability of generated recommendation results through the highlighting of certain areas of the product image as defined by the user based upon their previous user-generated review history.	"improves recommendations." Visual Explanation Quality: Quantitative metrics were likely used to assess how well the visual explanations matched the text reviews. User Perception (Qualitative): A human evaluation study was conducted to determine if the highlighted regions (ROIs) were actually relevant and interpretable to human users.	
(Azad et al.,2023)	Exploring Customer Behavior and Revenue Generation via Explainable Artificial	The Author developed an all-encompassing Explainable AI (XAI) system by combining various types of machine learning models (including Random Forests and Artificial Neural Networks) with interpretability	Confusion Matrix, Precision, Recall, and AUC	Online Shoppers Purchasing Intention Dataset

	Intelligence (XAI) in E-Commerce	frameworks (such as LIME and SHAP). After integrating the new Explainable AI (XAI) layer into existing e-commerce platforms, the Author found that this process greatly improves predictive performance while enabling users to understand which features of their purchasing intent process (e.g., PageValues and ExitRates) contributed most to their final purchase decisions.		
(Pleskach et al.,2023)	An E-Commerce Recommendation System Based on Consumer Behavior Models	The author has demo how the recommender system impacts the purchasing behavior based on the location and weather. It shows that during rainy day , the engagement have increase 14.2%. However, based on the study revealed ,Furthermore, the findings revealed distinct differences in the customer's perceived value between various areas of Ukraine where Odesa ranks the highest for CLV	Average Purchase Value: Tracks the average monetary value of transactions. Purchase Frequency Rate: Measures how often customers return to make a purchase.	Ukrainian E-Commerce Purchase Data

		confirming that models developed using best practices and the microservice architecture are superior to traditional approaches.	Customer Value: Used to identify high-value customer segments. Sales Impact: Measures the direct percentage increase in sales/participation triggered by external factors.	
(Vijayaraghavan and Mohapatra et al.,2023)	Stability of Explainable Recommendation	According to the authors, All feature-based explainable recommendation systems have similar weaknesses to external attacks; the authors demonstrated that the accuracy of their explanation drops dramatically as the level of external noise increases. The authors showed that adversarial noise causes more degradation than random noise because it optimises perturbations against the model's objective function. The sensitivity of models was also found to be affected by dataset size; the authors determined that models using explicit factor approaches are most sensitive on smaller datasets and neural networks exhibit the greatest drop in	NDCG: Checks if the best items appear at the top. Hit Rate: Counts if a relevant item appears in the list. Precision@K: Measures how many recommended items were actually relevant.	Amazon Electronics and Yelp

		performance on larger datasets. These results highlight the need for stronger defences against possible attacks.		
(Rahman et al.,2024)	Explainable AI-Driven Recommender Systems in E-Commerce	According to the author, when using Explainable AI (XAI) features, trust (\$M=4.1\$) and perceived transparency (\$M=4.3\$), as well as user distrust, were significantly higher than when using traditional black-box AI methods, which had lower levels of trust (\$M=3.2\$) and perceived transparency (\$M=2.9\$). It was also revealed through mediation analysis that increased trust and transparency brought about by XAI features were the main contributing factors to users making product purchase decisions, and that providing concise explanations decreased user distrust. Levels of trust and transparency were the primary drivers that positively influenced the users' actual decision to purchase products, proving that concise explanations effectively reduce user distrust.	User Trust: This metric measured the level of confidence users had in the system Perceived Transparency: This metric evaluated how clear and understandable the system appeared to users. Decision to Purchase: This behavioral metric tracked whether the user actually chose to buy the recommended item.	Primary Data
(Akhtar et al.,2024)	Explainable AI for Seller Recommendations (XAI-SR)	A hybrid recommender system, which integrates Matrix Factorization and Graph Neural Networks, is suggested by the author to	Model Accuracy: Comparative accuracy against baseline models User Trust: Percentage of	Real world E-Commerce Datasets

		improve both accuracy and usability through the use of XAI methods such as Integrated Gradients and DeepLIFT. The hybrid model produces far more accurate recommendations than either technology by itself. It has also been shown that providing users with an explanation for their recommendations leads to greatly increased confidence in the user's decision-making process; 82% of users indicated they would trust the hybrid recommender based on the explanation provided and 76% would make a purchase based on the recommendation. However, the author does suggest that there are challenges in terms of computational complexity as well as dealing with unstructured input data for the hybrid model.	survey respondents who trusted the explanations (82%). Decision Confidence: Percentage of respondents who felt confident making a purchase choice (76%).	
(Waykar et al.,2024)	Human-AI Collaboration in Explainable Recommender Systems	The author said if integrating the human collaboration into the EXPLAINABLE RECOMMENDER SYSTEM,it will increase user confidence and empowerment compared to traditional black box.However, the findings of this investigation suggest	Explanation Quality: Measures how well the explanation justifies the recommendation . User Understanding: Assess if the	

		that there is an important trade-off present; the level of detail provided in explanations must be balanced with how much they confuse the user by being overly complicated or too simplified, resulting in a lack of comprehension. In addition, without a uniform method for evaluating XRSs, researchers are not able to objectively compare the quality of the various types of explanation provided to different user populations.	user actually comprehends the logic behind the system's choice. Trust & Satisfaction: Evaluates the users confidence in the system and their happiness with the results. User Engagement: Tracks how actively the user interacts with the feedback loop and explanation settings.	
(Peya et al.,2025)	Machine Learning Based Potential Customer Recommendation System for E-Commerce Using Feature Selection and XAI	The Authors developed a recommendation system for smart prospective customers that uses both the Select K-best with Mutual Information (MI) feature extraction approach, along with multiple machine learning techniques to determine high-value customers based upon their grocery purchasing activities. The analysis of the recommendation system demonstrated that it could create	The authors created a rigorous evaluation framework that used both traditional classification metrics (accuracy, precision, recall, and F1) and the ROC-AUC score to evaluate the predictive power of the system. The results showed that the Random Forest	Consumer behavior and shopping habits dataset

		successful recommendations with 100% accuracy based upon a customer's purchasing history as determined by their model parameters. The Random Forest machine learning model demonstrated the greatest performance across all of the other methods tested. Furthermore, the authors were able to improve the interpretability of the generated recommendation results using Local Interpretability Model-agnostic Explanations (LIME) as a means to expose the effects of certain transactions and consumer habits on the results produced by the model.	method scored 100% in all of these categories and greatly exceeded regular methods such as FP-Growth and PCA. To confirm that the logic behind the model was aligned with the business objectives, LIME was also applied for qualitative validation in the form of feature importance weights and visualizations. Each of the components of their multilayered evaluation scheme was designed to build merchant confidence by assuring that the automated recommendations were clear and trustworthy.	
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2.6 Summary Of Chapter

The first part of this chapter describes the evolution of e-commerce recommender systems, addressing the processes of information filtering and the critical shift from

predictive accuracy to algorithmic transparency. In explaining the core techniques, Collaborative Filtering, Content-Based Filtering, and Hybrid architectures are evaluated in terms of their advantages, such as scalability and serendipity, and their limitations, particularly the "black-box" nature of deep learning models and the cold-start problem.

The chapter also provides an overview of existing literature regarding XAI, demonstrating that including XAI, such as using visual interpretability techniques or graph database recommendations for sellers, improves recommendation effectiveness and overall user experience for many online retail platforms. This establishes a framework for future chapters that will discuss how the e-commerce recommendation systems developed in this study employ XAI.

Chapter 3: Theoretical Framework

This chapter provides an overview of the theory behind an Explainable AI (XAI) E-Commerce Recommendation System, combining contemporary machine learning approaches with XAI, and detailing the three theoretical pillars on which the Recommender System rests. These are Content-Based Filtering to create recommendations, to extract features from the data, and XAI methods to produce interpretable models.

3.1 Operational Framework

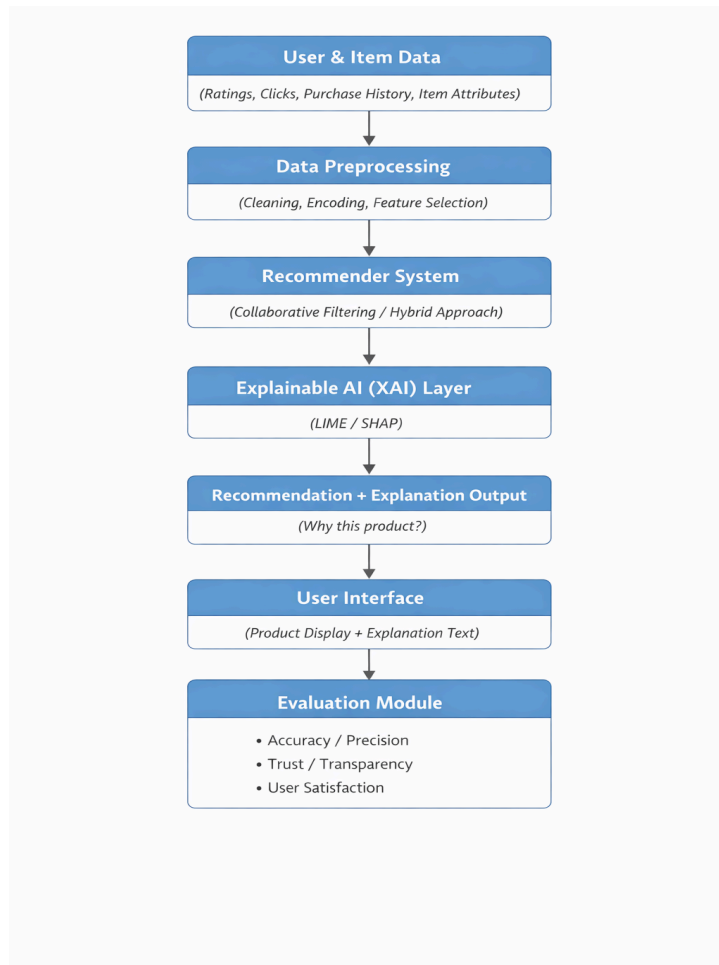


Figure 1.1

This section describes the workings of our Explainable AI-enhanced E-commerce Recommendation System and how each element interacts with one another to generate product recommendations for users and also provide users with explanations of why those recommendations were made to them.

In order for a model to generate recommendations, a dataset is collected from e-commerce websites and the information is grouped into three categories: user data, item data, and user-item interactions. User-item interactions are used to generate a personalised recommendation for each user.

The next step in our process is to prepare data for use in our recommendation system. We will identify any records that are missing values and remove any duplicate values from our dataset. Any categories in our dataset will also be

converted into a numeric format if they contain numbers or letters. Feature selection may also be applied to identify which attributes we want to include in our recommendations and explanations.

When the data has been cleaned up, it will be ready for training our recommendation engine. The engine will provide personalized recommendations of products based on the user's preferences. In order to do this, the system utilizes other algorithms, such as collaborative filtering and hybrid recommendation systems. By analyzing the historical interactions between users and items and the similarities between items, the algorithms will be able to determine how likely it is that a user will be satisfied with a given item based solely on the user's previous interactions. A ranked list of recommended items for each user will provide us with the output of this step.

After generating the recommendations, Explainable AI (XAI) enhances the recommendations with an explanatory component that provides an explanation of why the recommendation was made. To explain each recommendation, the use of post-hoc explanation methods (including LIME and SHAP) allows the system to identify the most relevant features or user behavior associated with the recommendation.

For instance, when a user receives a recommendation for a product, the system may explain that the recommendation was made because the user had previously purchased similar items. In addition to that, it may show that other users who made the same selection also purchased the same product or products with similar preferences.

The recommendation information and the corresponding explanation are shown to users through a user interface (UI) that displays the two concurrently. The product recommendation information and product explanation (of why that particular product was recommended) are displayed together for easy access. The UI is also designed to provide short, clear and direct explanations for each recommended product. Examples of explanation types include brief text descriptions and highlighted features. By offering the user these types of short, simple and concise explanations,

they will have a better understanding of the reasons for receiving that recommendation and will have greater confidence in the recommendation.

Lastly , the evaluation module for performance evaluation of the system contains measurement of recommendation quality based on the common evaluation metrics for similarity to users and on the metrics of accuracy and precision. The recommendation quality will also be evaluated from the point of view of the user using a user-centered evaluation approach, where users' perceived trust, perceived transparency, and user satisfaction will be measured, which may be done via questionnaires or feedback forms. The final evaluation results from the module will be evaluated to determine the degree to which the incorporation of Explainable AI improves the overall recommendation performance and user experience.

3.2 Natural Language Processing Techniques

3.2.1 Text Preprocessing

Text preprocessing is a crucial step in preparing textual data for machine learning models. The system implements the following preprocessing pipeline:

```
Raw Text → HTML Removal → Special Character Removal → Lowercasing → Tokenization → Stop Word Removal
```

Preprocessing Steps:

Step	Description	Purpose
HTML Tag Removal	Removes HTML markup using regex patterns	Clean text extraction
Special Character Removal	Removes non-alphanumeric characters	Noise reduction
Lowercasing	Converts all text to lowercase	Case normalization
Whitespace Normalization	Removes extra whitespace	Text standardization

3.2.2 TF-IDF(Term Frequency-Inverse Document Frequency)

TF-IDF is a numerical statistic that reflects the importance of a word in a document relative to a collection of documents. It is widely used in information retrieval and text mining.

Mathematical Formulation:

$$\text{TF-IDF}(t,d,D)=\text{TF}(t,d)\times\text{IDF}(t,D)$$

where:

- Term Frequency (TF): Measures how frequently a term occurs in a document where

$$\text{TF}(t,d)=\text{Total number of terms in } d/\text{Number of times } t \text{ appears in } d$$

- Inverse Document Frequency (IDF): Measures how important a term is across the corpus

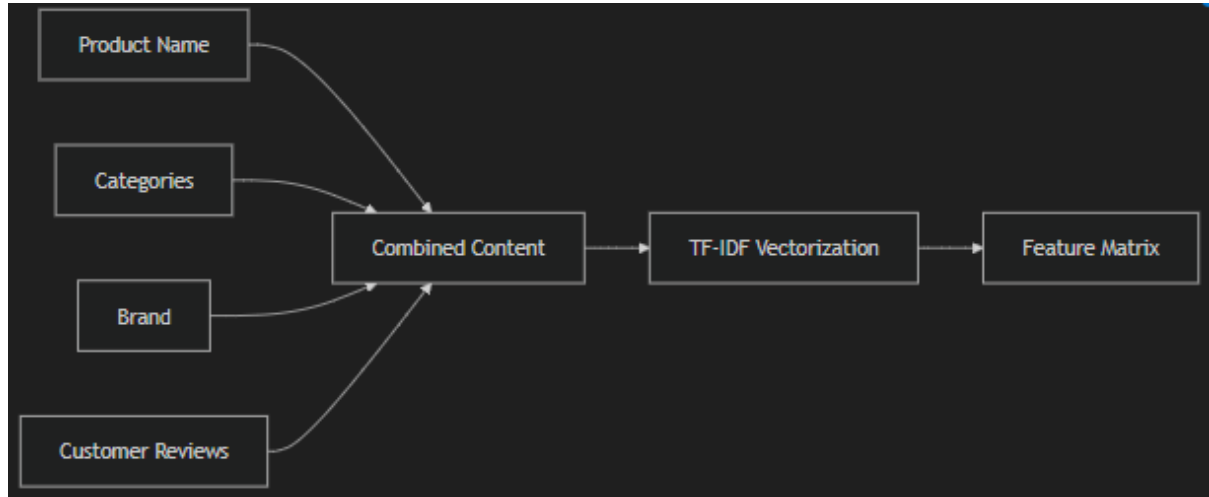
$$\text{IDF}(t,D)=\log\left(\frac{|D|}{|\{d\in D:t\in d\}|}\right)$$

Implementation Parameters in this Project:

Parameter	Value	Rationale
max_features	5,000	Balances vocabulary size and computational efficiency
ngram_range	(1, 2)	Captures both unigrams and bigrams for richer semantics
min_df	2	Filters out extremely rare terms
stop_words	'english'	Removes common English stop words

3.2.3 Feature Engineering

The system creates a comprehensive content representation by combining multiple text sources:



This multi-source approach ensures that the feature representation captures:

- . Product Identity: Through product names and brand information
- . Categorical Context: Through category hierarchies
- . User Perspective: Through aggregated customer reviews

3.3 Similarity Measures

3.3.1 Cosine Similarity

The purpose of cosine similarity is to measure the similarity between product feature vectors. It will calculate the angle of cosine between the two non zero vectors in a multidimensional space.

Mathematical Definition:

$$\text{cosine_similarity}(\mathbf{A}, \mathbf{B}) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

Properties:

- Range: $[-1, 1]$ for general vectors; $[0, 1]$ for TF-IDF vectors (non-negative)
- Scale Invariance: Unaffected by the magnitude of vectors
- Interpretability: Higher values indicate greater similarity

3.3.2 Similarity Matrix Computation

The system pre-computes a similarity matrix S where entry S_{ij} represents the cosine similarity between products i and j :

$$S = \text{TF-IDF Matrix} \times \text{TF-IDF Matrix}^T$$

This approach enables efficient of similar products in $O(n)$ time after $O(n^2)$ preprocessing step.

3.4 Evaluation Metrics

3.4.1 Root Mean Square Error(RMSE)

RMSE is a widely used metric for evaluating the accuracy of predicted values against the actual value. In the recommender systems, RMSE can measures how is the best the system can predicts user rating.

Mathematical Formulation:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Where:

- n represented the total number of observations,
- Y_i represents the actual prices for i -th observation,
- $\hat{Y}_i$ represents as the price predicted by the model of the same observation.

Interpretation:

Lower RMSE = Better prediction accuracy

RMSE is in the same unit as the ratings

3.4.2 Mean Absolute Error (MAE)

MAE provides an average magnitude of errors in predictions without considering direction.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Comparison: RMSE vs MAE:

Metric	Sensitivity to Outliers	Interpretation
RMSE	High (squares errors)	Penalizes large errors more
MAE	Low (absolute errors)	Treats all errors equally

3.4.3 Application in this Project

The system uses a weighted neighbor approach to predict ratings:

1. Find k similar products based on cosine similarity
2. Compute weighted average of their ratings:

$$\hat{r}_p = \frac{\sum_{j=1}^k s_{pj} \cdot r_j}{\sum_{j=1}^k s_{pj}}$$

Where:

- s_{pj} = Similarity between product p and neighbour j
- r_{pj} = Rating of neighbour product j

3.5 Summary of Chapter

The theoretical basis for Explainable Artificial Intelligence for E-Commerce recommendation systems has been outlined in this chapter. Several essential components combine together to create an effective and transparent product recommendation. The fundamental recommendation method is Content-Based Filtering which uses inherent characteristics of items to recommend similar products without consideration of user action history. In addition, TF-IDF Vectorization converts product attributes such as description, category and reviews into numeric feature representations. The next logical step is to calculate product similarity and rank them based on their similarity through the use of the Cosine Similarity metric. Two model are used to provide interpretability of product recommendations within the system; SHAP and LIME provided technical explainability of individual recommended items, while LLM-based Explanations provide natural language of why a product was recommended to non-technical users. Chapter 4 will describe how this theoretical framework will be implemented in a real-world setting, and Chapter 6 will report on the experiments conducted to evaluate its accuracy, providing the supporting evidence for the conceptual framework presented in Chapter 3.

Chapter 4: Research Methodology

4.1.1 Hardware and Software Requirements to Develop the Prototype

In this section, we define the hardware and software needed to build and implement the prototype for the Explainable AI E-Commerce Recommender System. The tools and resources described below provide a full range of capabilities required for processing data, developing models, displaying explanations, and evaluating the developed systems.

Hardware Requirements

The prototype does not require specialized or high-end hardware, as the system is designed for experimental and academic purposes. The minimum recommended hardware specifications are as follows:

- Processor (CPU): Intel Core i5 or equivalent (or higher)
- Memory (RAM): Minimum 8 GB RAM (16 GB recommended for smoother data processing)
- Storage: At least 256 GB of available disk space for datasets, libraries, and project files
- Graphics Processing Unit (GPU): Not mandatory. The prototype can be executed using CPU-based computation. However, a GPU may improve performance if deep learning models are extended in future work.
- Operating System: Windows 10 / macOS / Linux (Ubuntu)

Software Requirements The development of the prototype relies on widely used open-source tools and libraries to ensure reproducibility and ease of implementation.

Programming Language

- Python (version 3.8 or above)
Python is selected due to its extensive support for machine learning, recommender systems, and Explainable AI libraries.

Development Environment

- Jupyter Notebook or Visual Studio Code (VS Code)
Used for data exploration, model development, and experimentation.

Core Libraries

- NumPy & Pandas – Data preprocessing and manipulation
- Scikit-learn – TF-IDF vectorization, cosine similarity computation, and evaluation metrics (RMSE, MAE)
- Plotly – Interactive data visualization and result analysis
- Streamlit – Web application framework for the dashboard interface
- Requests – API integration for LLM-based explanations

Explainable AI Libraries

- LIME (Local Interpretable Model-agnostic Explanations)
Used to generate local explanations for individual recommendations.
- SHAP (SHapley Additive exPlanations)
Used to measure feature importance and provide consistent explanations of model predictions.

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